CODE HILLS 1001 — Business Plan

Terra Twine — Sisal (Agricultural Input)

> Part of the CODE HILLS 1001 business-plan library (6 plans total, built from Rwanda's natural resources: Umuravumba, Tea, Coffee, Umwenya, Sisal). Figures in RWF are labeled estimates — verify current local prices, supply arrangements, and regulatory requirements (RDB/RFDA/technical institute) before investing.

LOW-CAPITAL BUSINESS 3 — "Terra Twine" (Sisal — agricultural input)

1. Business Identity

- **Name:** Terra Twine
- **Tagline:** *"Grow more, waste less plastic."*
- **Resource:** Sisal.
- **Concept:** Hand-processed sisal twine, trellis cord, and biodegradable seedling sleeves sold to horticulture farmers and greenhouse operators as a direct substitute for plastic twine and plastic nursery bags, aligned with Rwanda's plastic-bag policy environment.
- **What makes it unique:** This is not a basket or craft business. It is an **agricultural input business** solving a specific horticulture production problem (plastic dependency in trellising/nurseries), sold B2B to growers and agro-dealers, not to tourists or gift buyers.

2. Problem

Rwanda's horticulture sector (tomatoes, passion fruit, cucumbers, and other trellised/staked crops, plus tree and vegetable nurseries) commonly relies on plastic twine and plastic nursery sleeves for training vines and starting seedlings. Rwanda's well-known restrictions on single-use plastics create both a compliance concern and a genuine market gap for a reliable, biodegradable, locally made substitute. **[ASSUMPTION — VERIFY LOCALLY: confirm current specific regulations applicable to agricultural plastic film/twine, as rules differ from those covering plastic shopping bags, before marketing on compliance grounds.]**

3. Solution

Terra Twine hand-twists sisal fiber into agricultural-grade twine and cord, and weaves simple biodegradable seedling sleeves, sold in bulk to greenhouse/horticulture farmers and agro-dealers as a direct, drop-in replacement for plastic equivalents — with the pitch centered on **production reliability and biodegradability**, not craft appeal.

4. Product

- **Trellis Twine (rolls, various lengths):** For tomato, passion fruit, and cucumber training.
- **Seedling Sleeves:** Woven sisal-fiber sleeves for tree/vegetable nursery starts, replacing plastic sleeves, which farmers currently must strip off and discard as waste before transplanting.
- **Variations:** A heavier-gauge cord for fencing/support structures.
- **Format:** Sold by the kilogram/roll to agro-dealers and directly to larger horticulture farms; no retail packaging needed.
- **Why customers want it:** Comparable functional performance to plastic for the growing season, biodegrades in soil (no plastic-strip removal labor at transplanting), and supports farms wanting to market their own produce as more sustainably grown.
- **Hard to confuse with a generic sisal product** because it is sold as a **functional agricultural input** through agro-dealer and farm-supply channels, with performance specifications (tensile strength, degradation time), not as a decorative craft item through gift/tourism channels.

5. Sustainability

- **Sourcing:** Sisal grown as hedgerow/boundary plants by smallholders in relevant districts; the business can pay small per-kilogram rates to households already growing sisal as a boundary plant, giving them new cash value for an existing planting.
- **Waste minimization:** Short fiber offcuts from processing are used as mulch or sold to Low-Capital Business 1 as packaging pouch material, or composted.
- **Packaging:** None needed for bulk twine rolls; seedling sleeves are themselves the "packaging" and are fully compostable.
- **Energy/water:** Manual decortication (fiber extraction) and twisting; no significant water or energy inputs at small scale.
- **End-of-life:** Fully biodegrades in soil within a growing season, unlike plastic twine/sleeves which persist and must be manually stripped and discarded.
- **Path to zero-waste:** As the business scales, offcuts and short fibers can be aggregated for the composite-panel input stream used in Premium Business 3, creating a two-tier internal resource flow (long fiber → twine, short fiber → composite panels).

6. Target Customers

- Horticulture/greenhouse farmers (tomato, passion fruit, cucumber).
- Tree and vegetable nurseries.
- Agro-dealers stocking farm inputs.

- Cooperative farms needing bulk trellising material.

7. Competitive Advantage

- **Advantage source:** A distribution and sourcing network (smallholder sisal hedgerows + agro-dealer relationships) combined with a genuine regulatory/production-cost tailwind (plastic-input pressure) that a craft/basket competitor does not tap into at all.
- **Why competitors may not immediately copy this:** It requires framing sisal as an **engineering/agricultural-input material** with performance claims, not a craft material — a positioning leap most generic “sisal basket” ideas never make, plus building agro-dealer B2B relationships rather than a market craft stall.

8. Startup Requirements

Essential: - Access to sisal leaves (own plants, or purchase from households with hedgerows). - Simple hand decorticating tool or method (e.g., scraping fiber from leaves using a fixed blade setup) — no powered machinery required at small scale. - Twisting/spinning jig (can be built from wood and a hand crank). - Storage/drying space for fiber. - Basic scale for bulk sales. **Add later:** - A small motorized decorticator once volume justifies it. - A delivery vehicle for bulk agro-dealer drop-offs. - Formal quality testing (tensile strength) via a local technical institute partnership.

9. Startup Budget (RWF, estimates)

Item	Minimum	Recommended
Sisal leaves (first batch)	15,000	40,000
Hand decorticating tool	10,000	30,000
Twisting jig (built)	10,000	25,000
Drying/storage space setup	0 (existing space)	20,000
Scale	8,000	15,000
Registration/branding	10,000	20,000
Initial agro-dealer sample stock	15,000	35,000
Total	**≈68,000–98,000**	**≈185,000**

10. Unit Economics

- **Twine (per kg):** Cost ≈ 900 RWF → Sold to agro-dealer at 1,800 RWF/kg → Gross profit ≈ 900 RWF (~50% margin).
- **Seedling sleeve (per 100-unit bundle):** Cost ≈ 6,000 RWF → Sold at 12,000 RWF → Gross profit ≈ 6,000 RWF (~50% margin).
- At **10 kg-equivalent units/day**: ≈9,000 RWF/day gross profit.
- At **30 units/day**: ≈27,000 RWF/day gross profit.
- At **100 units/day** (requires hired labor for decortication/twisting): ≈90,000 RWF/day gross profit before added labor costs, which become significant at this volume.

11. Production Process

1. Harvest mature sisal leaves.
2. Decorticate (scrape away pulp to expose fiber) by hand tool.
3. Wash and sun-dry fiber.
4. Twist/spin into twine using the jig; weave sleeves by hand.
5. Grade by strength/thickness; bundle for sale.
6. Quality-check for consistent thickness and no rot/mold before dispatch. **Quality/safety notes:** Store dried fiber away from moisture to prevent mold before sale; wear hand protection during decortication (fiber edges can cause cuts/irritation).

12. Marketing Strategy (first 30 days)

- **Week 1:** Visit 5 agro-dealers with free twine samples and simple spec sheet (length, thickness, indicative tensile strength).
- **Week 2:** Approach 2–3 horticulture farms directly for a free trial roll during their current trellising season.
- **Week 3:** Collect farmer feedback (did it hold up through the season so far); document with photos for future marketing.
- **Week 4:** Secure first paid agro-dealer stocking order.

13. Sales Strategy

- **Where:** Agro-dealers (primary channel), direct sales to larger horticulture farms (secondary), cooperative bulk orders.
- **Pricing:** Priced close to plastic twine equivalents, sometimes slightly higher, justified by biodegradability and no stripping labor at harvest.
- **First 10 customers:** Local greenhouse/horticulture farmers known personally or through a cooperative contact.
- **Feedback:** End-of-season check-in on twine durability.
- **Model:** Primarily B2B bulk/contract sales to agro-dealers, supplemented by direct farm sales; move toward standing seasonal supply contracts once trust is established.

14. Growth Plan

1. **Home production** — hand tools, single household.
2. **Small workshop** — hired part-time fiber processors, dedicated drying/twisting space.
3. **Local brand** — recognized agro-input brand stocked by multiple district agro-dealers.
4. **Rwanda-wide distribution** — supply agreements with national agro-dealer networks/cooperative unions.
5. **Regional/export opportunity** — supply to similar horticulture sectors in neighboring EAC countries facing comparable plastic-reduction pressure — **[ASSUMPTION, unverified, longer-term]**.

15. Risks

Risk	Mitigation
Fiber supply inconsistency	Multiple household suppliers; own small sisal planting as backup
Manual processing bottleneck	Plan for a motorized decorticator once volume justifies the investment
Quality variation affecting trust	Simple consistent grading and testing before every batch dispatch
Competition from cheap plastic twine	Emphasize total-cost-of-use (no stripping labor, no disposal problem)
Seasonal horticulture demand	Diversify into fencing/general cordage uses in off-season
Regulatory claims risk	Avoid overstating “banned plastic” claims without verifying current rules

16. 90-Day Launch Plan

- **Weeks 1–2:** Confirm sisal access; build/borrow decorticating tool.
- **Weeks 3–4:** Produce first test batch of twine and sleeves.
- **Weeks 5–6:** Free trials with 2–3 farms; gather early feedback.
- **Weeks 7–8:** Approach agro-dealers with samples and spec sheet.
- **Weeks 9–10:** Register business; finalize pricing with first agro-dealer.
- **Weeks 11–12:** Fulfill first bulk order; set up basic bookkeeping; reinvest in fiber-processing capacity.

Originality Test

1. **Obvious idea:** Sisal baskets, mats, or decorative crafts sold to tourists.
2. **Structural difference:** An agricultural-input business (twine/seedling sleeves) sold B2B to horticulture farmers and agro-dealers, not a craft/tourism product.
3. **Distinctive combination:** Resource (sisal) + problem (plastic-twine dependency in horticulture) + customer (greenhouse/nursery farmers, agro-dealers) + positioning (functional ag-input with performance specs, not craft).
4. **Hard to copy because:** Requires agro-dealer B2B relationships, consistent quality grading, and a mental repositioning of sisal as an engineering material — none of which a craft-stall competitor is set up to do.

PART 2 — THREE PREMIUM BUSINESSES